

Network Technology Written Assignment

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Module: IT-1562-04

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Introduction:

The TCP/IP model is a communication model that allows information to transfer from one device to another through encapsulation and de-encapsulation processes. The TCP/IP model contains the following layers: Application layer, Transport layer, Internet layer and Network access layer. Each of the layers plays a crucial role in the transmission of the data so that the communication can work.

The TCP/IP communication process:

Encapsulation:

The full Encapsulation process starts from the Application layer of the device and ends at the network access layer. Intermediary devices will re-encapsulate what they de-encapsulate but with different data at the network access layer.

1. Application Layer:

At the Application layer, all the key information that the end devices want will be created and formatted into data which contains the protocol, for example, HTTP, that contains the request the user wants, where this data will later be encapsulated in the lower layers of the TCP/IP model.

2. Transport Layer:

The transport layer encapsulates the data with a TCP header, which contains the source port and destination port number of the source and destination devices depending on the type of protocol that the device requests, and other fields like sequence and acknowledge number. It can also segment the data into smaller sizes if the data is too big and reassemble the data when received. Having a TCP header in the data ensures that the data is delivered successfully. It is then passed down to the Internet Layer.

3. Internet Layer:

The IP header will encapsulate the data with an IP header, which contains the source and destination of the source and end devices and various fields so that when the intermediary devices receive the data packets, they will know where to forward the packet. The data is passed down to the Network access layer.

4. Network Access Layer:

This layer will encapsulate the data with a frame header and trailer, both sides contain different fields. The header includes the source and destination MAC address and other fields, and the frame trailer consists of the frame check sequence (FCS). After that, the data will then be transported to the destination via a media, which could be wired or wireless.

De-encapsulation:

The full de-encapsulation process usually happens when the data reaches an end device. For intermediary devices, different devices will de-encapsulate different numbers of layers that are necessary to function.

1. Network Access Layer:

The data is received from the media, the destination MAC address from the frame header is read, and the FCS is checked in the frame trailer. If the destination MAC address matches the current MAC address and nothing wrong in the FCS, it will de-encapsulate the header and trailer and pass it to the internet layer.

2. Internet Layer:

The internet layer will check the destination IP address from the IP header. If the IP header matches the device's, it will de-encapsulate the IP header before passing the data to the transport layer.

3. Transport Layer:

The transport layer will check if the port numbers in the TCP header match. They will also check other fields like sequence and acknowledge to make sure there are no errors. Once checked, it will de-encapsulate the header and pass it to the application layer.

4. Application Layer:

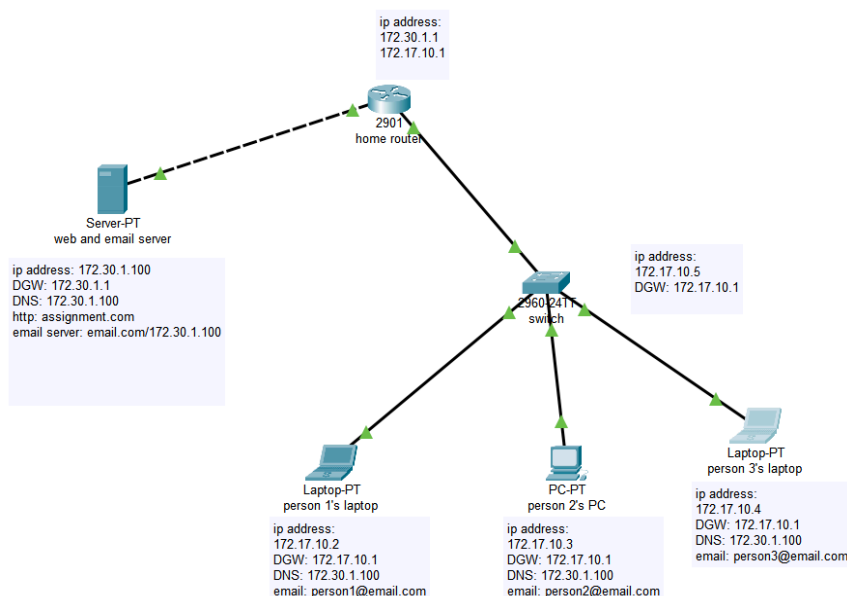
It will process and interpret the data received and give a response depending on the type of protocol (In the example above, HTTP) received. It will then send a response back to the source device using the same encapsulation process or display the HTTP in a presentable format to the user.

Network Setup:

Network Table:

Refer to Appendix 1

Image of Network Setup/Logical Topology:



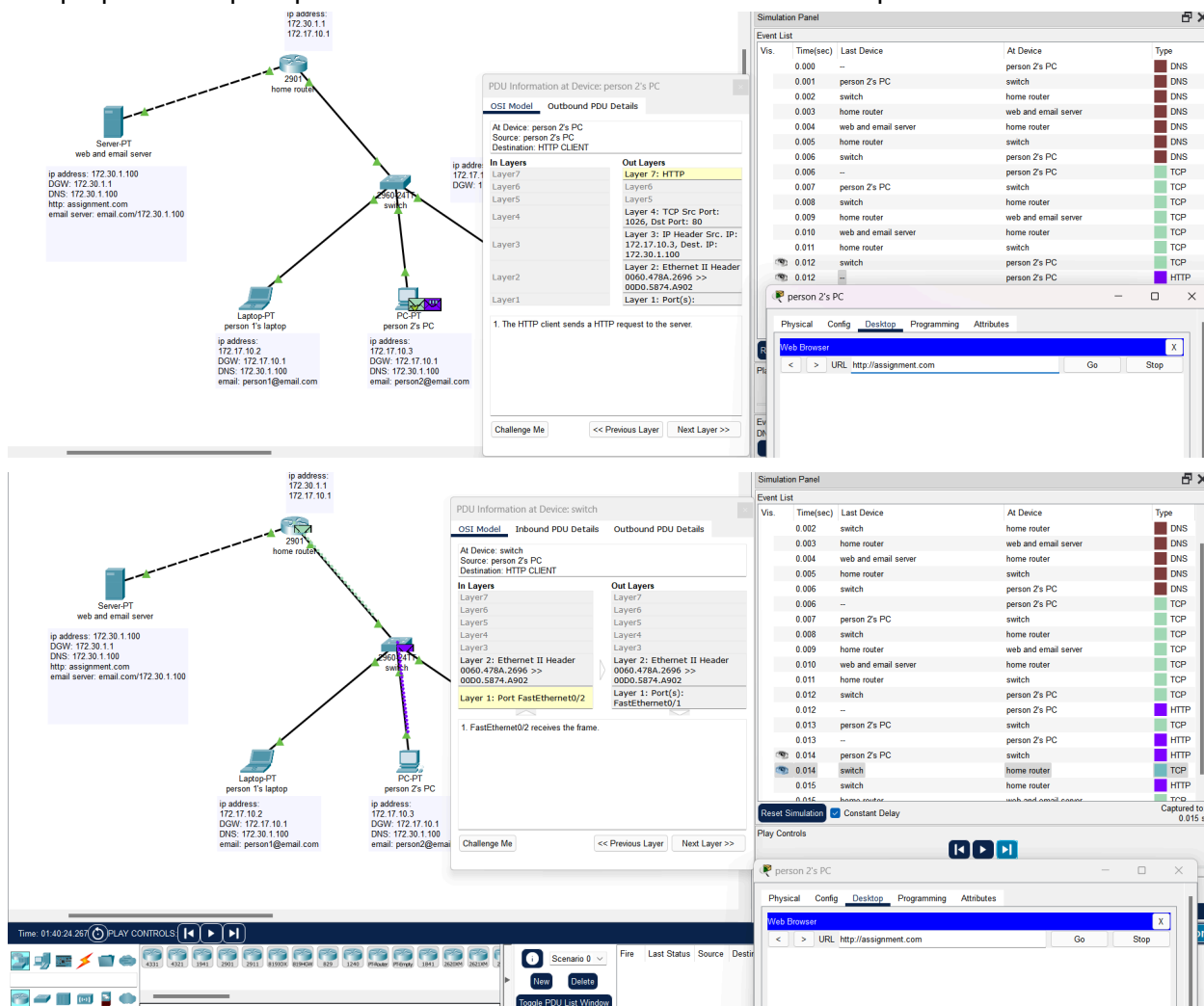
TCP/IP process and routing of the HTTP:

In this setup, I used person 2's PC to access the web and email server using the DNS, <http://assignment.com>, to access the webpage. The user only key in assignment.com into the search bar. After the server receives the DNS query and responds to the PC, the PC creates a TCP request to keep the connection on before creating an HTTP request.

Step 1: HTTP encapsulation in the PC.

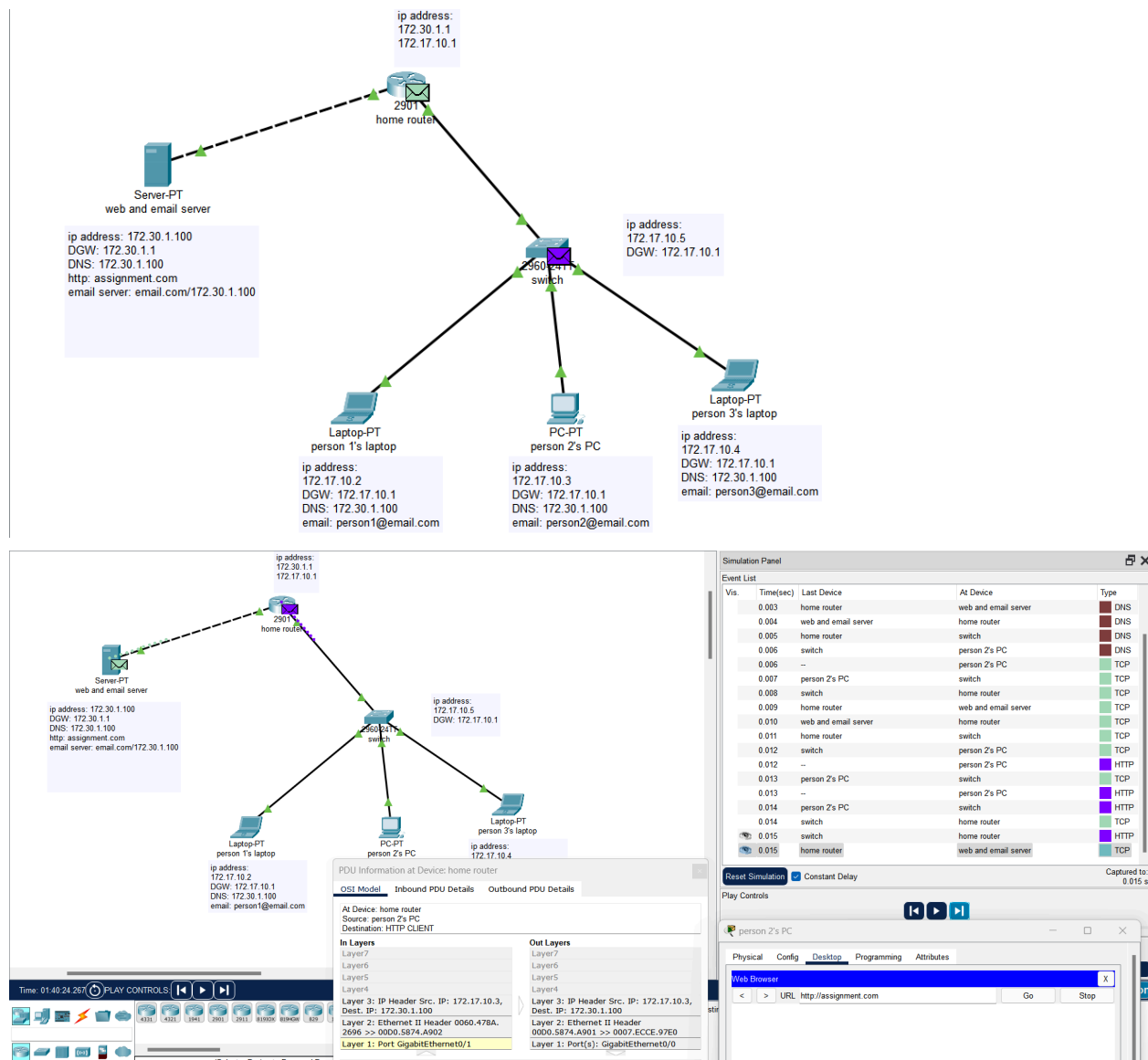
The HTTP data is then encapsulated with the TCP header, where the destination port number is 80 and the source port number is 1027, the IP header, which contains the source IP address (172.17.10.4) and destination IP address (172.30.1.100), the Frame header and trailer where it contain the source MAC address (0060.478A.2696) and destination MAC address (00D0.5874.A902). The data will then be passed to the switch via the straight-through wire that connects to the switch at port Fa0/4 from the PC's port Fa0.

The purple envelope represents the data which contains the HTTP request.



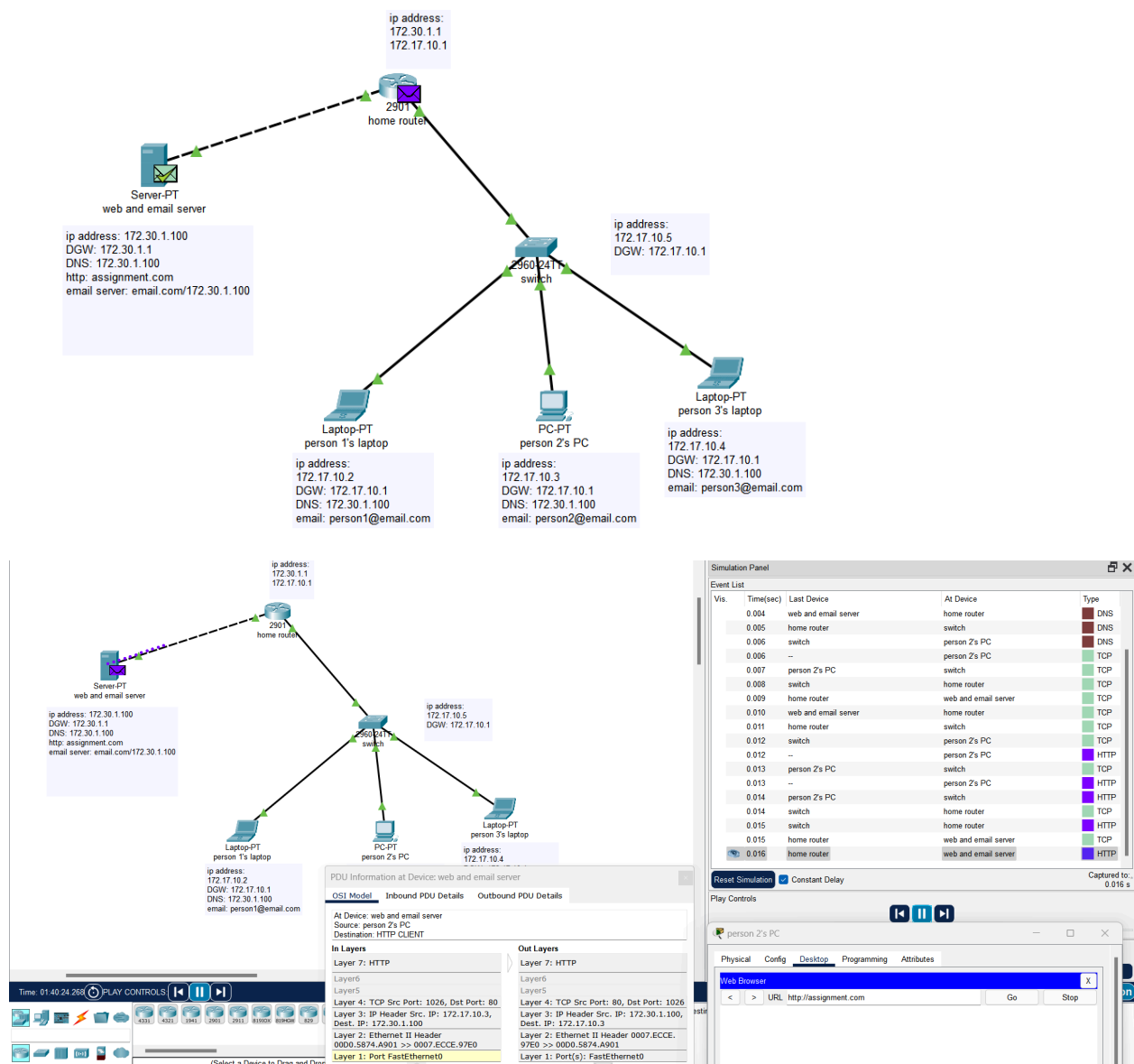
Step 2: At the switch

The switch receives the data and checks the destination MAC address using its MAC table. After that, it sends the data to the router without de-encapsulation and encapsulation of the data. The data is sent from the switch's port Fa0/1 to the router's port Gig0/1.



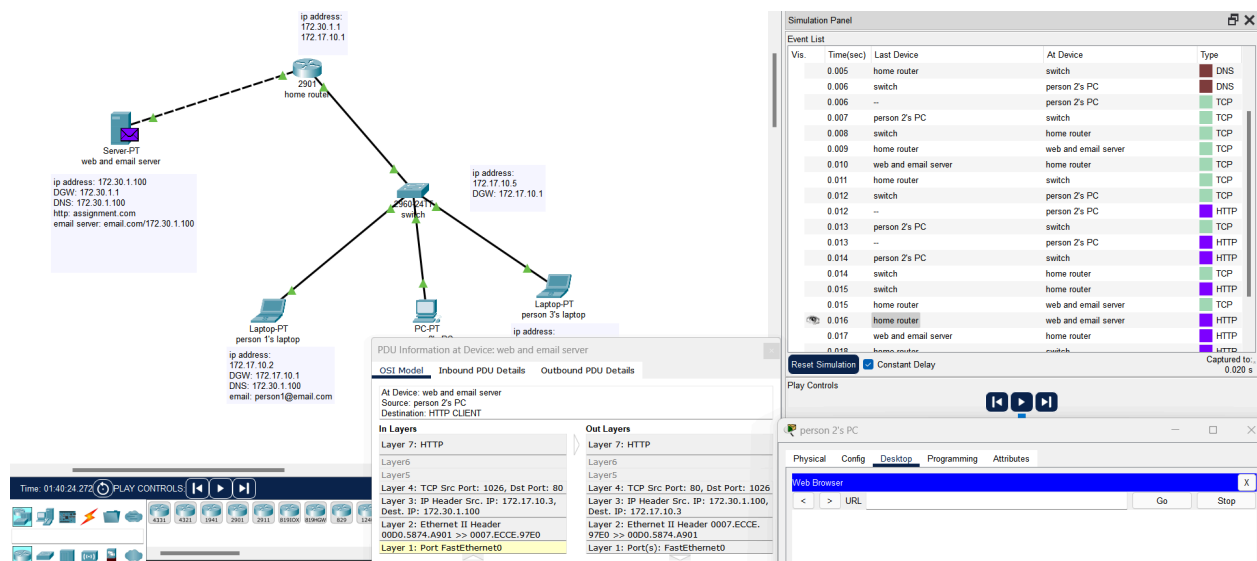
Step 3: At the Router

The router will check the destination MAC address before de-encapsulating the frame headers and trailers to check for the IP address. After that, The router will check for the IP address on its CEF table. It will encapsulate the data with a different frame header, with the source MAC address being 00D0.5874.A901 and the destination MAC address being 0007.ECCE.97E0. Then it sends the data to the server from the router's port Gig0/0 to the server's Fa0 port.



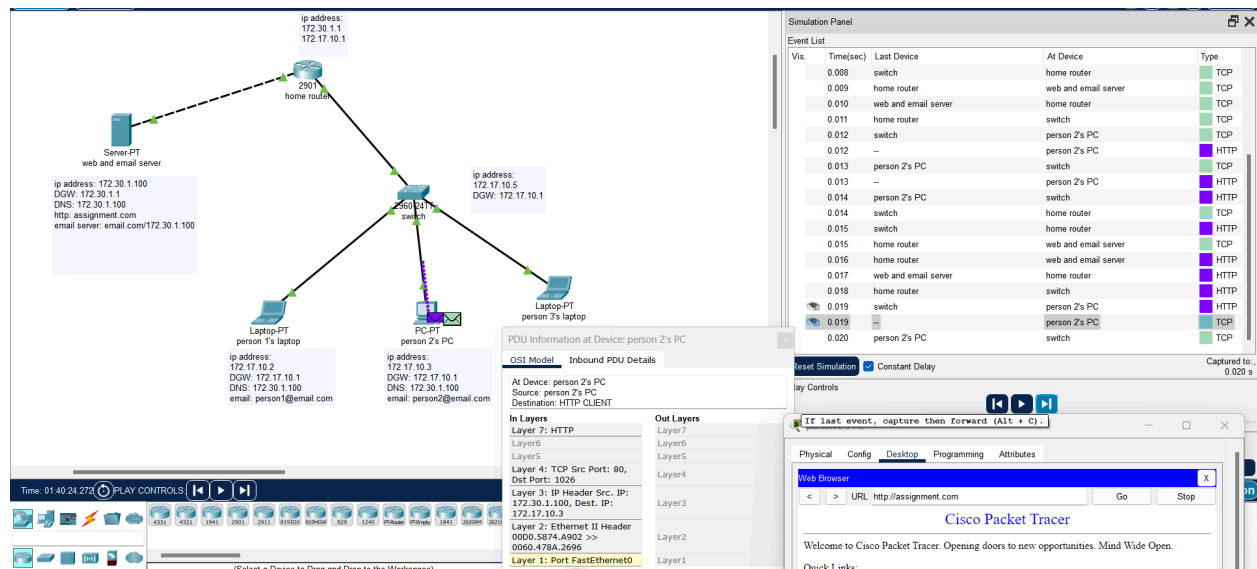
Step 4: At the Server

The server will check the destination MAC address in the frame header and the destination IP address in the IP header before de-encapsulating everything to find the HTTP request. After that, it will send an HTTP response that allows the PC to connect to the web page. It will have to encapsulate the response again, with the source IP address being 172.30.1.100 and the destination IP address 172.17.10.4. The source MAC and Destination MAC address will be the reverse of step 3 before it is sent back to the router, repeating the process again but in reverse, until it reaches the PC.



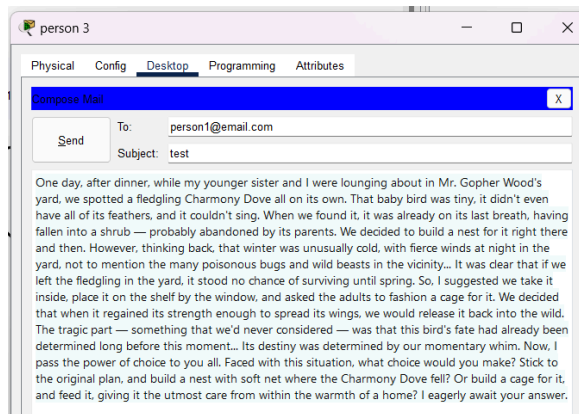
Step 5: Back to Person 2's PC

The PC can access HTTP after it receives the HTTP response through de-encapsulation, allowing person 2 to view the webpage.



TCP/IP process in Email:

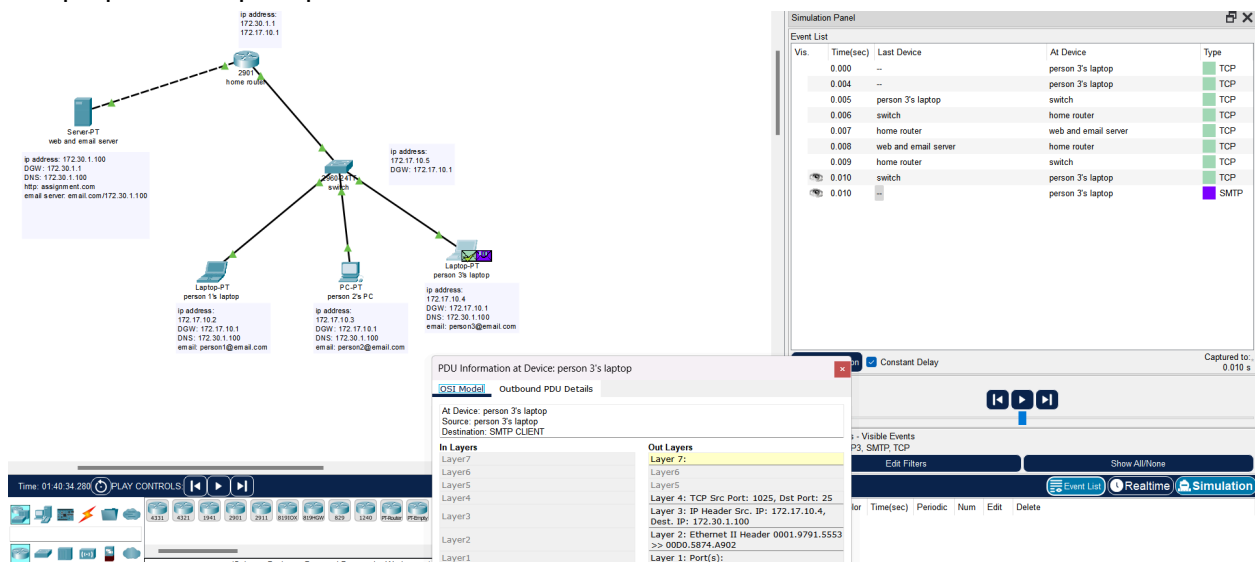
Person 3 wants to send Person 1 an email. Hence he typed what he wanted to send and pressed send. Person 3's email address is person3@email.com and person 1's is person1@email.com.

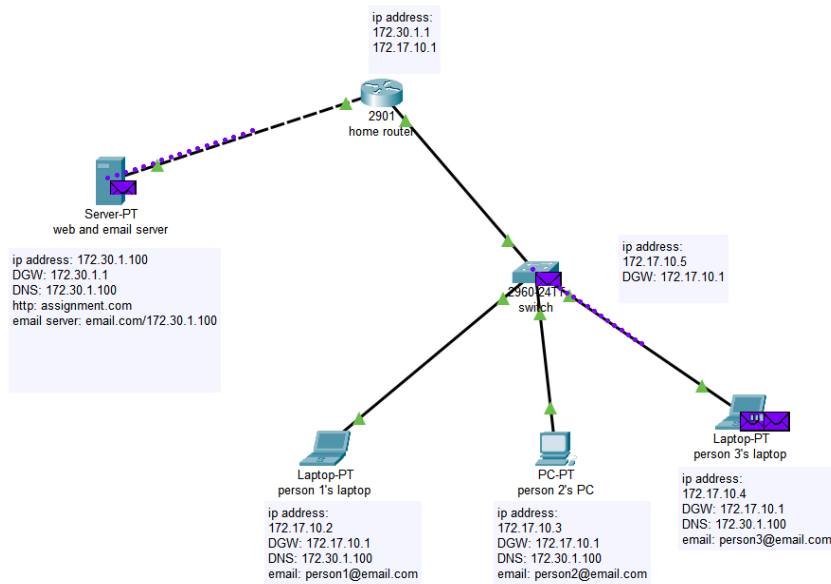


Person 1's laptop will first want to send a TCP request to the server for a TCP connection like what the above did, then after receiving the reply from the server and establishing a TCP connection.

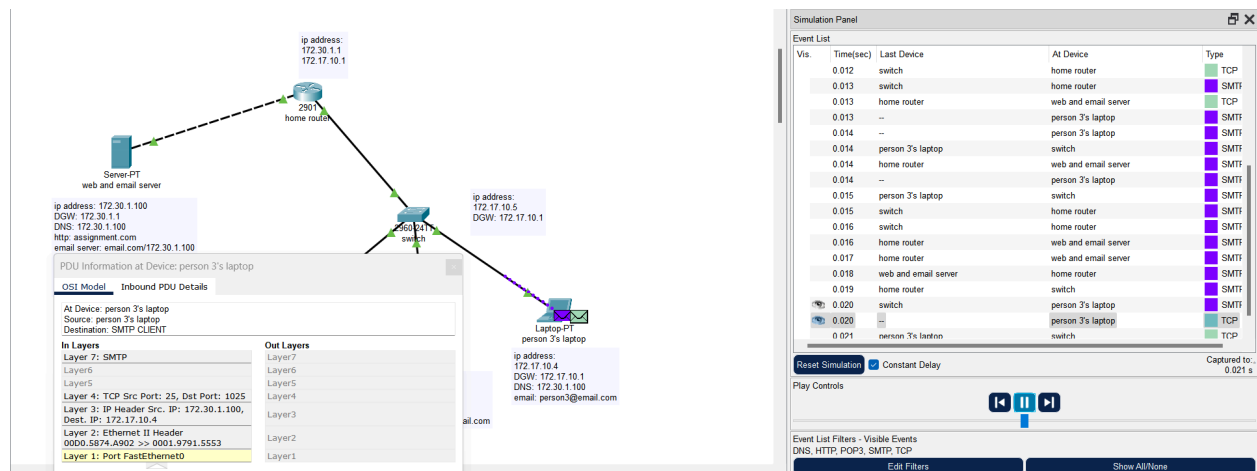
How the TCP/IP model supports email communication is that it will first encapsulate the SMTP data with the necessary headers and trailer. Since the email is a bit big, the transport layer will segment the data into smaller parts so that it will not be discarded due to exceeding the limit of 1518 bytes. Then it will encapsulate each of them with the TCP header, IP header, frame header and trailer. Then it proceeds to send the data one by one to the server. It essentially works the same as how the above.

The purple envelope represents data with SMTP



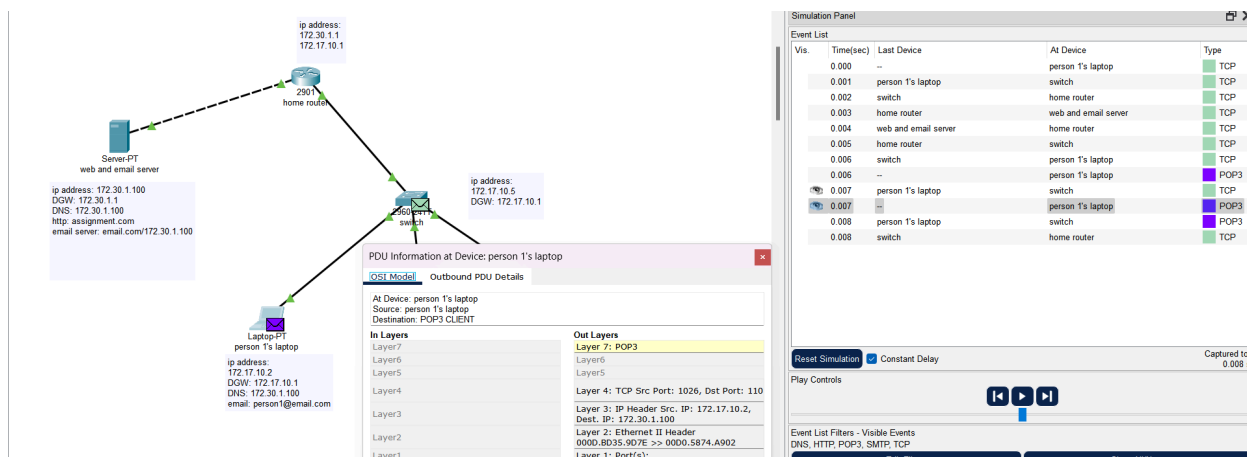


When there are no more segments to receive, the server will send back the SMTP data to the Laptop, which then prompts the laptop to send TCP data to close the connection.

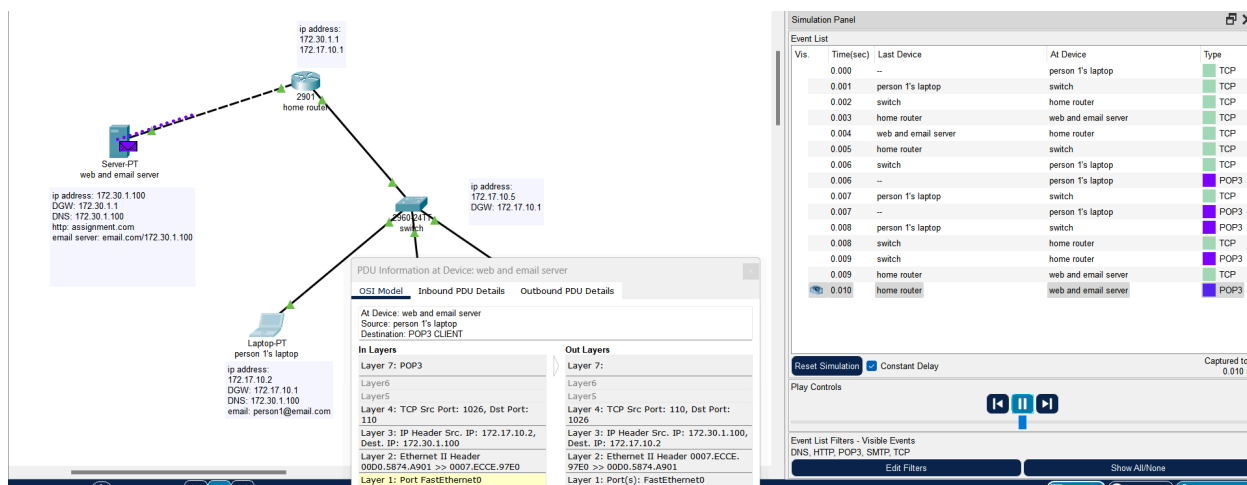


As for person 1, when he wants to receive the email, he presses the receive button, and he will send a pop3 request to the server after sending the TCP request. It is the same as how the TCP/IP model encapsulated the SMTP data with the necessary headers and trailer, but this time, there will be more traffic coming from the server to his Laptop as is as the email is already segmented into smaller parts. After receiving all the necessary data, it is later reassembled into the message Person 3 sent so that he can read the email.

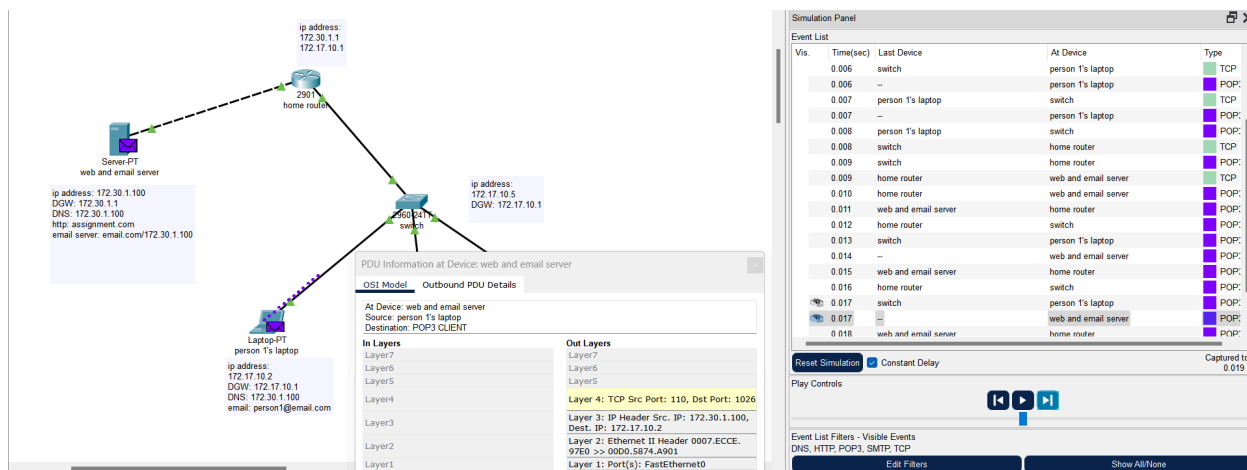
The purple envelope represents the POP3 data



(Laptop create request)



(request arrive at server)



(multiple pop3 response to laptop)

The main difference other than different source IP addresses and initial source MAC addresses is at the application layer, instead of using HTTP, they used SMTP and POP3, but the functions for encapsulations and sending of data remain the same.

Conclusion

The TCP/IP model is a great way to communicate between devices, as shown above, it supports many protocols for communication between devices. Each layer plays a part in ensuring the data is formatted correctly to ensure that the data reaches its destination accurately and reliably. Hence, the TCP/IP model has become one of the most common communication models to be deployed worldwide.

Citation and references

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Cisco Packet Tracer built-in explanation

Appendix:

Appendix 1:

Device	Port	Connected to which Switch port	IP Address	Subnet Mask	Default Gateway	MAC address	DNS
Person 1's laptop	Fa0	Fa0/3	172.17.10.2	255.255.255.0	172.17.10.1	000D.BD35.9D7E	172.30.1.100
Person 2's PC	Fa0	Fa0/2	172.17.10.3	255.255.255.0	172.17.10.1	0060.478A.2696	172.30.1.100
Person 3's laptop	Fa0	Fa0/4	172.17.10.4	255.255.255.0	172.17.10.1	0001.9791.5553	172.30.1.100
S2960-24 TT Switch	Fa0/1 Fa0/2 Fa0/3 Fa0/4	-	172.17.10.5	255.255.255.0	172.17.10.1	Fa0/1: 0090.215D.9501 Fa0/2: 0090.215D.9502 Fa0/3: 0090.215D.9503 Fa0/4: 0090.215D.9504	
2901 Home router	Gig0/0	Fa0/1	172.17.10.1	255.255.255.0		00D0.5874.A902	
2901 Home router	Gig0/1	-	172.30.1.1	255.255.255.0		00D0.5874.A901	
Web and Email server	Fa0	-	172.30.1.100	255.255.255.0	172.30.1.1	0007.ECCE.97E0	172.30.1.100